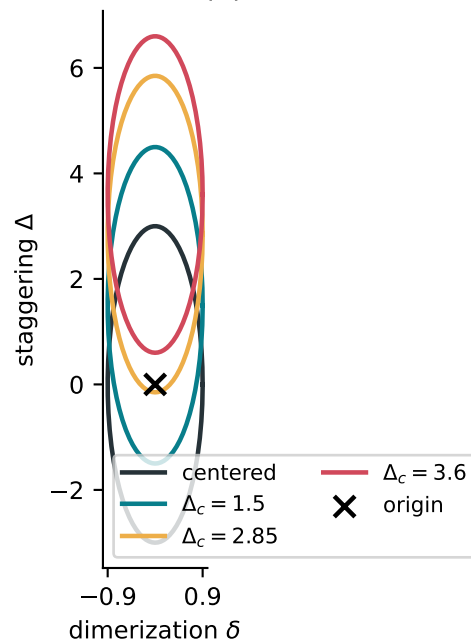
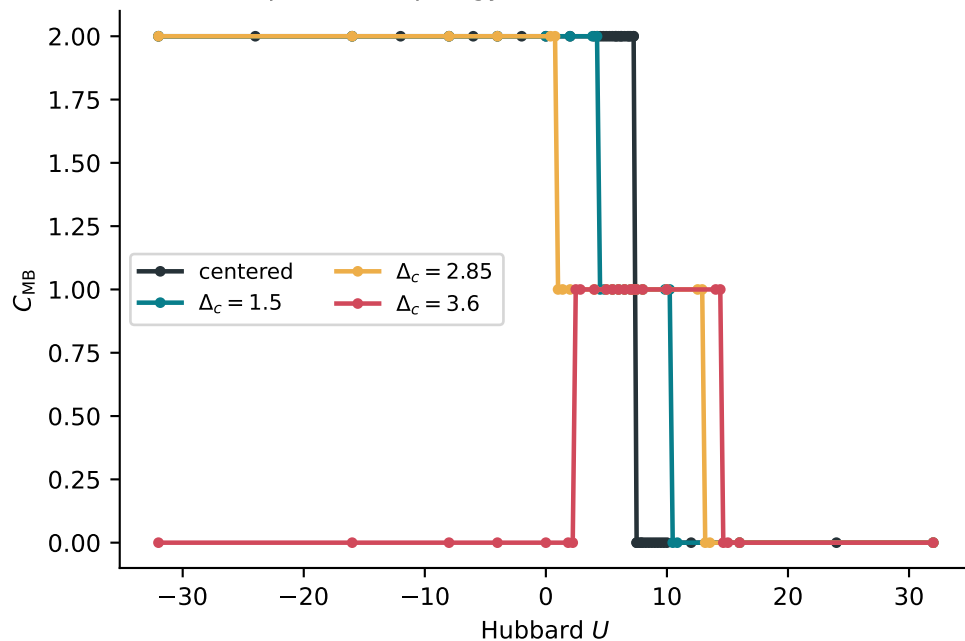


$L=8, t=1$, translated Rice--Mele pump paths

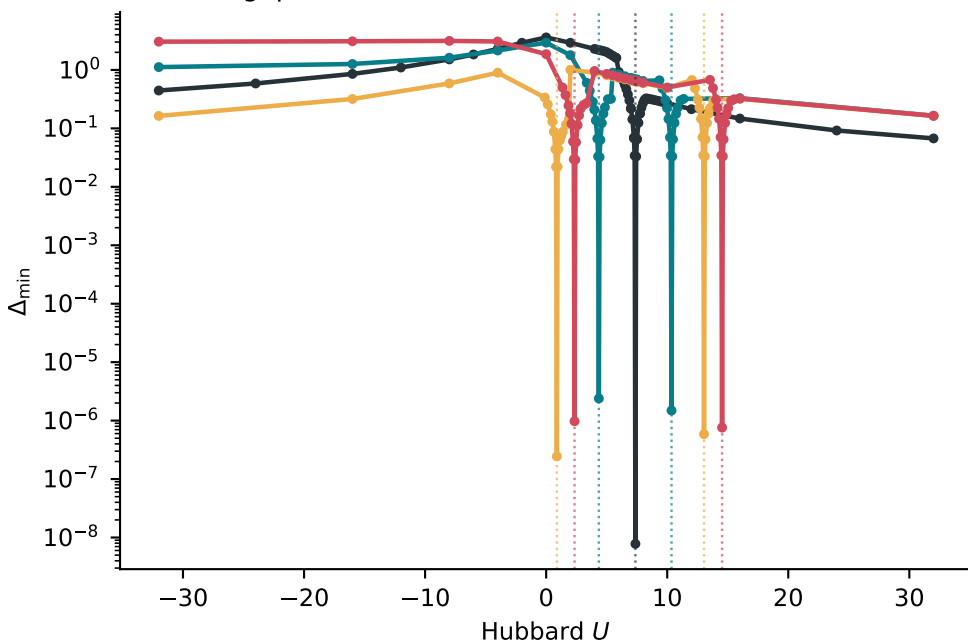
(a) Pump paths



(b) Path-dependent topology



(c) Torus gap with critical-line refinement



(d) Twist-averaged topological efficiency

